

Phase 5C CVRP Runbook

This runbook extends the replay-aware benchmark-transfer study from TSP/ATSP to real vehicle-routing benchmarks from CVRPLIB.

1. Prepare The Benchmark Pack

Activate the environment first:

```
conda activate Python3_14
python prepare_cvrp_benchmarks.py
```

That downloads the selected CVRPLIB subset, verifies official solution files against recomputed route cost, and writes benchmarks/cvrp/manifest.json.

2. Offline Smoke Test

```
python run_cvrp_suite.py --config configs/cvrp_suite/smoke_suite.json --skip-judge
```

Use this after changes to the CVRP engine, benchmark loader, or replay-selection logic. The smoke suite writes into DO NOT COMMIT/cvrp_suite/smoke because it is a verification artifact, not an official experiment run.

3. Main CVRP Extension Suite

```
python run_cvrp_suite.py --config configs/cvrp_suite/01_replay_transfer.json
```

The four default conditions are:

- cvrp_no_replay
- cvrp_random_replay
- cvrp_failure_replay
- cvrp_failure_replay_compression

Primary interpretation target:

- final held-out CVRPLIB optimality gap

Secondary targets:

- final synthetic holdout gap
- combined transfer gap
- code novelty
- heuristic complexity
- adaptation efficiency

Selection-time transfer probes combine held-out CVRPLIB probe instances with the adversarial routing layout pool. The exact synthetic holdouts remain reserved for final evaluation.

4. Replication

```
python run_cvrp_suite.py --config configs/cvrp_suite/01_replay_transfer.json --seed-offset 1000
--replicate-label b
```

Recommended minimum:

- 10 paired offsets for an official result

Official main-study target:

- 20 paired offsets using the fixed offset list 0, 1000, 2000, ..., 19000
- Use the same offset list for every condition and every routing family so the paired comparisons remain valid.

5. Aggregation

```
python aggregate_cvrp_runs.py --runs-root runs/cvrp_suite/replay_transfer
```

That writes:

- aggregate_summary.json
- aggregate_report.md
- aggregate_report.pdf
- paired bootstrap comparison tables for the replay conditions